BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE APPLICATIONS AND GEO-INFORMATICS

WEEKLY PROGRESS REPORT(08/06/2026 –12/06/2026)

WEEK 4

AUTOMATIC GEO-REFERENCING	**AUTOMATED GCP DETECTION AND GEO-TIFF GENERATION**

PROJECT DESCRIPTION : AUTOMATED GCP DETECTION AND GEO-TIFF GENERATION

GROUP MEMBER : Chirag Jindani, Vedant Pathak

GROUP ID : 2026G262

GROUP GUIDE: Dr. Manoj Pandya

GROUP COORDINATOR : Dr. Manoj Pandya

COLLEGE GUIDE NAME : Dr. Shweta Saharan

COLLEGE GUIDE No. : 9509676076

COLLEGE GUIDE EMAIL : shweta.saharan@gsv.ac.in

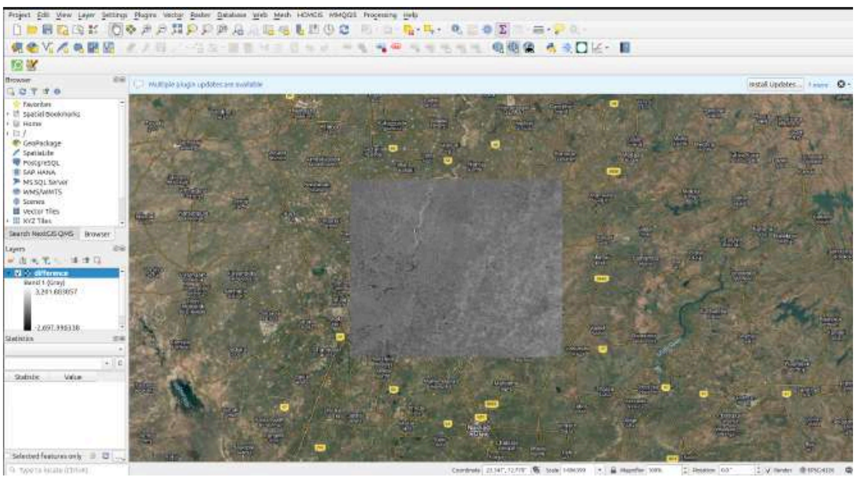
COLLEGE NAME : GatiShakti Vishwavidyalaya, Vadodara

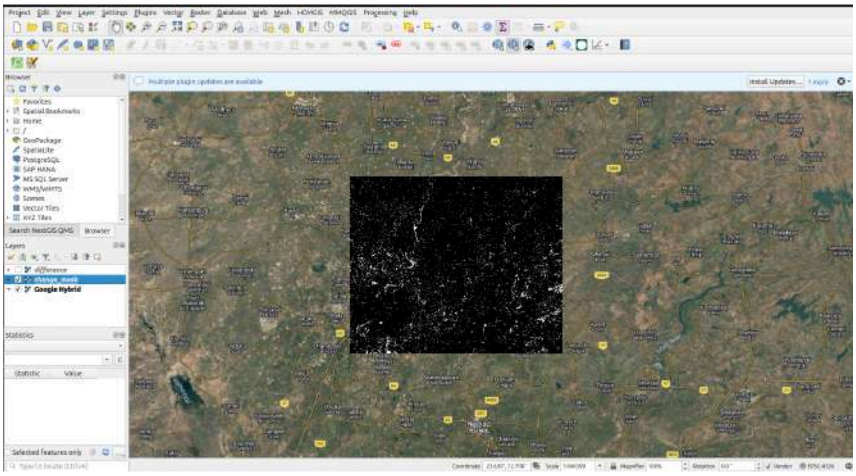
08/06/2026	**Containerised the full application stack (Next.js frontend and FastAPI backend) with github** **Performed end-to-end validation of the deployed georeferencing pipeline — capture, upload, feature matching, warping, and GeoTIFF export — confirming the project is fully functional in production.**
09/06/2026	**Added a new “Change Detection” tool to the application alongside the existing georeferencing workspace, with a toggle to switch between the two.** **Built the Change Detection dashboard with dual drag-and-drop upload zones for the ‘before’ (t1) and ‘after’ (t2) raster images, including previews and stage tracking.**
10/06/2026	**• Applied Otsu thresholding and morphological opening/closing to convert the** **SSIM difference map into a clean binary change mask.** **• Detected and filtered contours from the cleaned mask to isolate significant** **change regions, drawing annotated outlines on both input images.** **• Built the results view with zoom/pan controls, a side-by-side annotated image** **comparison, and summary metrics (change count, changed area, alignment** **score)**
11/06/2026	**Started a satellite-imagery change detection pipeline using the Google Earth Engine Python API, authenticating against a GCP project and defining an area of interest.** **Configured access to the Sentinel-2 harmonized surface reflectance collection, selecting Band 4 (Red) and Band 8 (NIR) across two separate time periods (2021 vs. 2026).** **Built a download routine that generates cloud-composited mean images for each period and retrieves them as GeoTIFFs via Earth Engine’s export API.**
12/06/2026	**Split the inspection layout into a coordinate data table**

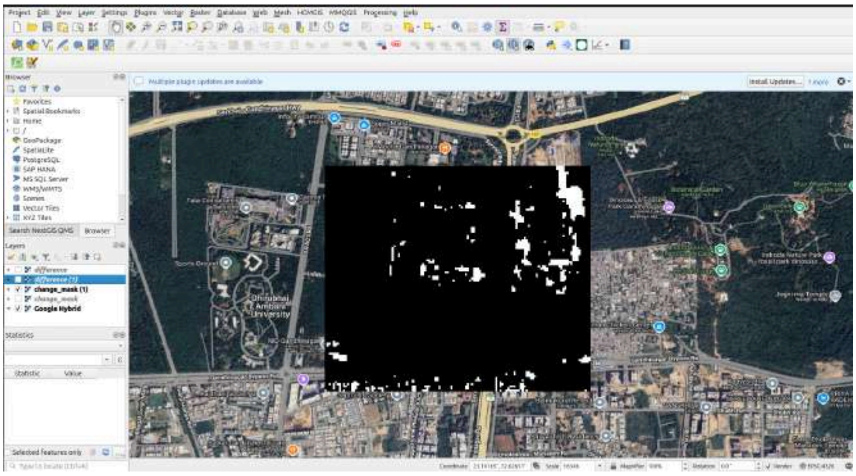
opposite the visual view workspace.

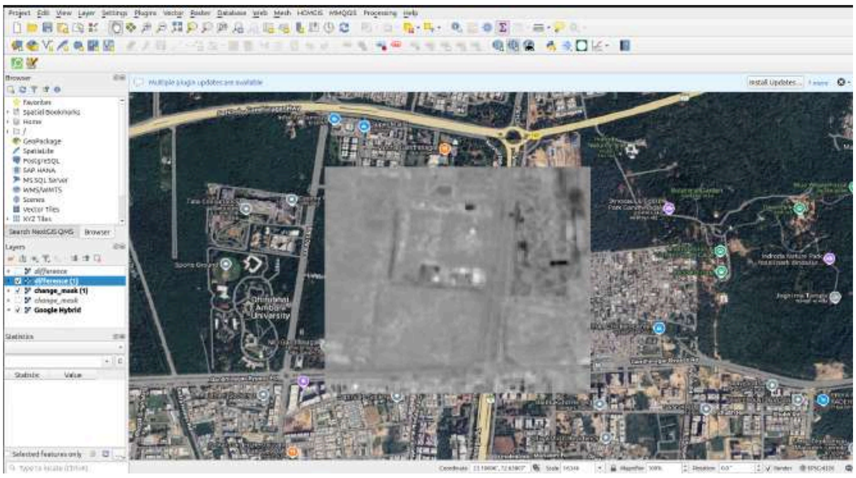
	Built custom math handlers managing smooth mouse drag-to-pan and scroll wheel-to-zoom actions. **Authored bounding constraints trapping the translation coordinates within a strict 30% out-of-frame limit.** **mplemented spatial alignment (reprojection) and histogram-based radiometric normalisation so the two time-period rasters are directly comparable.** **Implemented the MAD (Multivariate Alteration Detection) transform — canonical variate analysis with chi-square statistic computation — to identify genuine change pixels.** **Exported the resulting change-mask and difference layers as GeoTIFFs, converted the change mask into a GeoJSON polygon layer, and generated diagnostic visualisation plots**

Next Week Plan	During the final week, the focus will be on conducting the final project run and verifying the complete workflow. Thesis preparation, result validation, and final project submission will also be completed.

screenshots	







references:

https://earth.google.com/web